

Markelo — Assumption Map

Purpose: This document lists every important assumption in the Markelo PRD and user stories. Each assumption has a risk score and a question to ask in the field.

How to read the risk score:

- **Regret cost** = how much work we must redo if the assumption is wrong.
- **Confidence** = how sure we are today, based on real evidence.
- **Priority** = test the HIGH regret + LOW confidence rows first.

Important: Confidence values below are honest. The team has spoken to approximately 4 people (founder statement, 20 July 2026 meeting). Almost nothing here is validated.

Tier 1 — Test these first (high regret, low confidence)

A1. Institutions will attach a control sheet to every booklet before the exam

- **Source:** PRD A7 Feature 1, Workflow Step 3–4
- **Status:** ASSUMED
- **Why it matters:** This is the entry point of the whole product. Every later step depends on it.
- **The problem:** The PRD says Markelo requires "no change to the physical examination process." This is not accurate. Attaching a unique, per-student control sheet to each booklet **is** a change to exam-day process. Someone must print 1,240 sheets, keep them in order, and match the correct sheet to the correct student.
- **If wrong:** The script-identification pipeline (US-06, US-07, US-08) does not work. This is a large part of the product.
- **Ask:** "Walk me through what happens on exam day, from when booklets leave the store to when the invigilator collects them. Who handles them at each step?"

A2. Institutions own scanners that can bulk-scan thousands of pages

- **Source:** PRD Workflow Step 5 ("using existing scanners")
- **Status:** ASSUMED — never verified
- **Why it matters:** If a department has one flatbed office scanner, scanning 1,240 scripts is impossible.
- **If wrong:** The institution must buy hardware. This changes the price, the sales cycle, and the buyer.
- **Ask:** "What scanning or photocopying equipment does the department have? May I see it?" (Record make, model, and sheet-feed capacity.)

A3. Anonymous marking is wanted, and is permitted by institutional regulation

- **Source:** PRD A9 — described as "Markelo's foundational trust feature"
- **Status:** ASSUMED
- **Why it matters:** This is the core of the product and the main sales claim.
- **The risk:** Wanting anonymity is not the same as being allowed to have it. Some institutions may have exam regulations that require the marker to confirm identity. Some lecturers may resist it.
- **If wrong:** A large part of the product's value disappears.
- **Ask:** "When you mark, do you see whose script it is? Has your institution ever discussed hiding student identity? What is the written exam regulation on this?"

A4. The institution's existing result portal will accept a Markelo export file

- **Source:** PRD A7 Feature 7 ("a result file the institution's own portal already accepts")
- **Status:** ASSUMED — highest technical risk in the product
- **Why it matters:** If the export does not load, the exam officer must re-type every result. The main efficiency claim fails.
- **If wrong:** Markelo needs a per-institution integration. That is slow, costly, and blocks scaling.
- **Ask:** "What system do you upload final results into? Can you show me the file format it accepts? Who controls that system?"

A5. Scripts go missing often enough to be a real, felt problem

- **Source:** PRD A3 Problem Statement
 - **Status:** ASSUMED — this is the product's founding claim
 - **Why it matters:** If missing scripts are rare, the main pitch is weak.
 - **Ask:** "Think of the last two exam periods. Did any script go missing? What happened? How long did it take to resolve?" (Ask for the incident, not an opinion.)
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Tier 2 — Test soon (high regret, medium confidence)

A6. Teaching Assistants exist, and lecturers already delegate marking to them

- **Source:** PRD A8, Feature 4 — called "a first-class workflow, not a side feature"
- **Status:** ASSUMED
- **Risk:** TA availability differs a lot by institution and department. Some lecturers mark everything themselves. Some use postgraduate students informally, with no official record.
- **Ask:** "Who marks scripts for your largest class? How is the work shared? Is that arrangement official?"

A7. Markers have a device good enough for the marking interface

- **Source:** Implied by US-10, US-11

- **Status:** ASSUMED
- **Risk:** If lecturers mark on a phone, the whole marking screen design changes. The PRD assumes desktop web.
- **Ask:** "What device would you use to mark on screen? Do you have a personal laptop? Where would you do it — office or home?"

A8. Institutions will accept scanned scripts being stored on an external server

- **Source:** Implied throughout; PRD C6 mentions compliance
- **Status:** NEEDS VALIDATION — includes a legal question (Nigeria Data Protection Act)
- **Risk:** Exam scripts are sensitive records. An institution's legal or ICT team may refuse external storage, or require on-premise hosting.
- **Ask (ICT/legal):** "Where is student data allowed to be stored? Do you have a data-protection policy? Has an external system been refused before?"

A9. Extra answer sheets can be handled by one of two methods

- **Source:** PRD/US-03 (extra-sheet method 1 or 2)
- **Status:** ASSUMED
- **Why it matters:** Students who write more than one booklet are common. If the extra sheet is not linked correctly, pages join the wrong script. This is a silent, serious data error.
- **Ask:** "What happens when a student needs a second booklet? How is it joined to the first one today?"

A10. Continuous Assessment (CA) scores exist in a spreadsheet that can be merged

- **Source:** PRD A7 Feature 7
- **Status:** ASSUMED
- **Ask:** "Where do CA scores live now? Who holds them? Can you show me the file?"

Tier 3 — Business and buying assumptions (currently the weakest area)

A11. We know who signs the contract

- **Status:** UNKNOWN — this is a gap, not an assumption
- **Note:** The stated target users (lecturers, exam officers) are users, not buyers. Nobody has confirmed who holds the budget.
- **Ask (any senior staff):** "If a department wanted software like this, who approves it? Who pays — the department, the ICT directorate, or the central administration?"

A12. Institutions have budget, and a purchase cycle we can survive

- **Status:** UNKNOWN
- **Ask:** "When does the budget year run? How long does approval usually take for a new system?"

A13. Private universities are the right first target

- **Source:** Founder statement, 20 July 2026 meeting (they respond faster)
- **Status:** ASSUMED — the reasoning is about *speed of response*, not *willingness to pay or fit*
- **Risk:** Fast to reach is not the same as likely to buy. Federal universities have far higher script volumes, so the pain may be much greater there.
- **Ask:** Compare volume and pain between one private and one federal institution before fixing the target.

A14. Faster result release is valued enough to pay for

- **Source:** PRD A3
- **Ask:** "What is the deadline for submitting results? What happens if it is late? Has that ever caused a real problem?"

Tier 4 — Lower regret (design details, cheap to change later)

ID	Assumption	Source
A15	Moderation works by sampling a subset of scripts	PRD A8 Step 10
A16	The moderated score should become authoritative on approval	US-14 Business Rule
A17	A marker may correct their own marks until moderation starts	US-10 Business Rule (PRD flags the exact cutoff as unconfirmed)
A18	Exam officers can produce a clean roster spreadsheet	US-05
A19	5,000 roster records must validate within 60 seconds	US-05 BR-05.4
A20	Demo Mode reduces the need for training	PRD A10

How to use this map

1. Take Tier 1 and Tier 2 questions into every interview.
2. After each interview, mark each assumption: **Supported / Contradicted / Still unknown**.
3. Review the map weekly. Move rows between tiers as evidence arrives.
4. **Do not mark an assumption "validated" from one conversation.** Use three or more independent sources.